UNIT - V : REPORT WRITING & FIELD INSTRUMENTATION

Soil exploration reports | Strain gauges | Piezometers | Pressure cells | Inclinometers | Proving ring | Load cells | Displacement gauges

Two parts of this unit: (A) the soil exploration report - how to identify, calculate and present the findings; (B) field instrumentation - devices used to monitor the performance of soil, rock and structures during and after construction.

PART A - SOIL EXPLORATION REPORT

1. Purpose and importance

The report is the final product of a soil investigation. It communicates the subsoil conditions to the designer and contractor, and recommends the type and depth of foundation and allowable bearing pressure. It is also a legal document - claims for "differing site conditions" refer to it. (IS:1892-1979 "Code of practice for subsurface investigation for foundations").

Qualities of a good report

- Clear, concise, systematic - arranged in logical order with an index.
- Factual data separated from interpretation.
- Sufficient number of boreholes with plan, logs, sections.
- Gives design parameters (c, phi, gb, Es, cv) with ranges, not single numbers.
- States assumptions and limitations, date of W.T. observation.
- Signed and dated by a responsible engineer.

2. Identification - what must be identified?

Item	Identification
Project	Name, client, type of structure, column loads, floor levels, area of site, proposed foundation level

Item	Identification
Site	Location map (with lat-long), topography, vegetation, adjacent structures, drainage, flooding, previous land use (fill, tank, quarry)
Geology	Geological map, rock/soil origin, faults, earthquake zone (IS:1893), special soils (black cotton, laterite, collapsible, peat)
Soil strata	Layer-wise thickness, type, group symbol, colour, consistency/density, special features
Ground water	Depth, date, fluctuation, flow direction, chemical analysis
Samples & tests	Locations, depth, quantity, standards (IS:2720 parts) followed

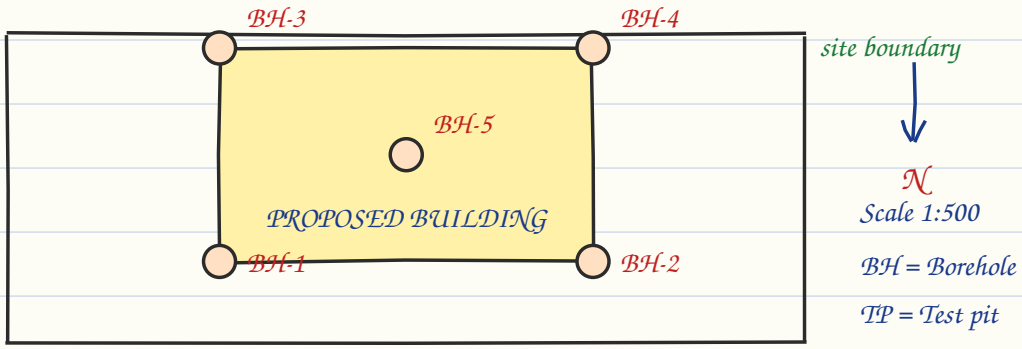


Fig 5.1 Borehole location plan

3. Contents and format of the report

Section	Contents
1. Title page & index	Project name, report no., date, consultant, list of contents/figures/tables
2. Summary	One-page: soil profile, W.T., recommended foundation, SBC, major risks
3. Introduction	Scope, terms of reference, structure details, standards used
4. Field work	Boring method, equipment, depth of each BH, sampling, SPT, in-situ tests
5. Lab work	Tests: water content, G, grain size, Atterberg, shear (UCS/ triaxial/ direct), consolidation, CBR, chemical
6. Subsoil description	General profile, individual layer description, W.T.
7. Analysis	Bearing capacity, settlement, pile capacity, slope stability etc.
8. Recommendations	Foundation type, depth, allowable pressure, precautions, dewatering, filling, retaining wall

Section	Contents
9. Annexures	Bore logs, test data sheets, graphs, photographs, drawings, borehole location plan, soil profiles

4. Calculations in the report

4.1 Index properties summary

Tabulate for each sample: depth, w, gd, G, Sr, e, wL, wP, Ip, % gravel/sand/silt/clay, IS classification; and strength: c, phi (or cu), compressibility: Cc, cv, e0.

4.2 Safe bearing capacity (Terzaghi / IS:6403)

Terzaghi (square footing, general shear): $q_u = 1.3\,c\,N_c + \gamma\,D_f\,N_q + 0.4\,\gamma\,B\,N_\gamma$; strip: $q_u = c\,N_c + \gamma\,D_f\,N_q + 0.5\,\gamma\,B\,N_\gamma$; circular: $1.3\,c\,N_c + \gamma\,D_f\,N_q + 0.3\,\gamma\,B\,N_\gamma$.

Net ultimate $q_{nu} = q_u - \gamma\,D_f$. Net safe $q_{ns} = q_{nu} / F$ (F = 2.5 to 3). Safe $q_s = q_{ns} + \gamma\,D_f$.

Local shear failure (soft/loose): use $c' = 2c/3$, $\tan\phi' = (2/3)\tan\phi$.

Example 1. Square footing B = 2 m at D_f = 1.5 m in clay (phi = 0): c = 40 kPa, γ = 18 kN/m3 ; N_c = 5.7 (Terzaghi, phi=0), N_q = 1, N_γ = 0.

$q_u = 1.3 \times 40 \times 5.7 + 18 \times 1.5 \times 1 = 296.4 + 27 =$ 323.4 kPa

$q_{nu} = 323.4 - 27 = 296.4$; $q_{ns} = 296.4 / 3 =$ 98.8 kPa ; safe pressure = 98.8 + 27 = 125.8 kPa.

Example 2 (sand from SPT, Meyerhof). Corrected N = 20, B = 2 m (> 1.2 m), D_f = 1.5 m, permissible settlement 25 mm.

$F_d = 1 + 0.33\,D_f/B = 1 + 0.33 \times 0.75 = 1.25$ (limit 1.33)

$q_a = 11.98\,N\,[(B + 0.3)/B]^2\,F_d = 11.98 \times 20 \times (2.3/2)^2 \times 1.25 = 11.98 \times 20 \times 1.3225 \times 1.25 =$ 396 kPa (check against shear failure and W.T. correction).

4.3 Settlement

- **Immediate:** $S_i = q B (1 - \nu^2) I_f / E_s$
- **Consolidation:** $S_c = C_c H/(1 + e_0) \times \log_{10}[(\sigma_0' + \Delta\sigma)/\sigma_0']$ for a normally consolidated clay layer thickness H.
- **Time:** $t = T_v H_{dr}^2 / c_v$ ($T_v = 0.848$ for 90 % consolidation).
- **Allowable:** isolated footing 40 mm (sand), 65 mm (clay); raft 75 mm; differential 1/500 to 1/300 (IS:1904).

> **Example 3.** 4 m thick NC clay: $e_0 = 0.9$, $C_c = 0.3$, $\sigma_0' = 80$ kPa at mid-depth, $\Delta\sigma = 40$ kPa. $S_c = 0.3 \times 4/(1.9) \times \log_{10}(120/80) = 0.632 \times 0.1761 = 0.1112$ m = 111 mm.

4.4 Foundation recommendation logic

Soil condition	Recommended foundation
Good bearing soil within 1.5 - 3 m	Isolated / strip footing
Weak soil near surface, firm below 5 - 6 m	Raft, pile or under-reamed pile
Black cotton / swelling soil	Under-reamed piles, sand cushion, belled piers, CNS layer
Very soft clay/ high W.T.	Pile / pier, ground improvement (stone columns, preloading)
Rock within 3 m	Footing on rock; rock bolt / anchor if required
Fill / loose sand liquefiable	Compaction, piles to firm layer, remove/replace

5. Preparation of the report - step by step

- Collect all field logs and lab data; check and reconcile (logs vs sample vs test results).
- Group data layer-wise, prepare idealised soil profile and design values (use average/ lower bound).
- Draw BH location plan, subsoil sections, graphs (N vs depth, grain-size, consolidation curves).

- Carry out analysis for foundation type as above (with factor of safety) and check settlement.
- Write the recommendations with clear statements (depth of footing from which G.L., pressure, precautions).
- Review by senior, final printing (colour photographs, drawings at legible scale), signing.

PART B - FIELD INSTRUMENTATION

Geotechnical instrumentation is used to measure pore pressure, total stress, load, strain, settlement, lateral movement - to (i) verify design assumptions, (ii) control construction (rate of filling, staged loading), (iii) warn of failure, and (iv) improve future designs (observational method, Peck).

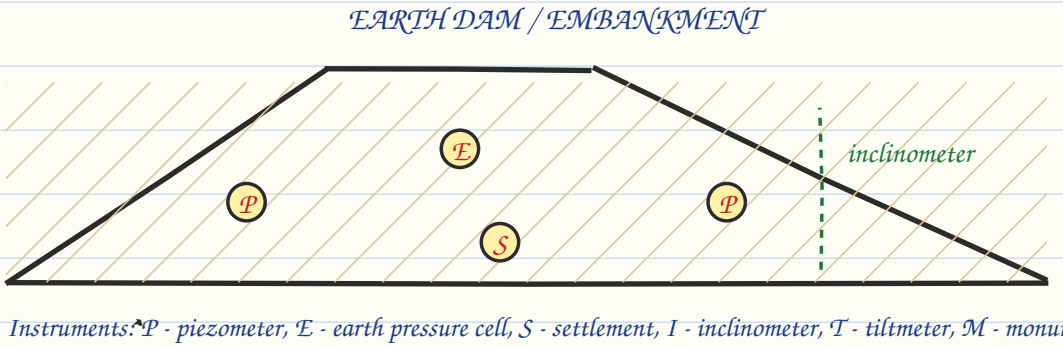
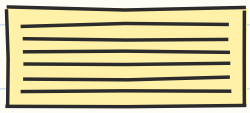


Fig 5.2 Instrumentation of an earth dam

Quantity to measure	Instrument
Pore water pressure / ground water	Piezometers (standpipe, pneumatic, vibrating wire)
Total earth pressure	Pressure cells (earth pressure cells)
Strain in steel / concrete	Strain gauges
Lateral movement	Inclinometer, tiltmeter
Settlement, vertical movement	Settlement gauge, extensometer, level survey
Load in anchors, struts, piles	Load cells, proving rings
Displacement, deformation	Dial gauge, LVDT, crack meter, tape extensometer

6. Strain gauges

Strain gauge (foil type):



grid of Constantan wire

GF = (DR/R)/e (gauge factor ~ 2.0)

e = DL/L strain; stress s = E e

Types: electrical resistance (bonded, foil, weldable)

vibrating wire: e = K(f2 - f0^2)

Quarter / half / full Wheatstone bridge

Fig 5.3 Electrical resistance strain gauge

- Principle: change in length -> change in electrical resistance of the conductor: DR/R = GF x e.
- Bonded foil gauges are glued onto steel/concrete; connected in Wheatstone bridge (temperature compensation by dummy gauge). Reading unit = strain indicator in microstrain.
- Vibrating wire strain gauge: wire in tension; plucked; frequency f measures strain: e = K(f2 - f0^2). Stable for years; used in dams, piles, tunnel linings.
- Embedment gauges (in concrete) and spot weldable gauges (on steel / pile / strut).
- Applications: stresses in pile/ struts / tunnel lining, load tests on piles, rock bolts, retaining wall reinforcement.

Type	Merits	Demerits
Electrical resistance	Cheap, small, good for dynamic	Drift, moisture sensitive, cable length affects
Vibrating wire	Long-term stable, long cables	Not for fast dynamic events, costlier
Mechanical (Demec)	No electronics	Manual, lower accuracy

7. Piezometers

Measure pore water pressure (u) - essential for effective stress analysis and stability of embankments, dams, slopes and excavations.

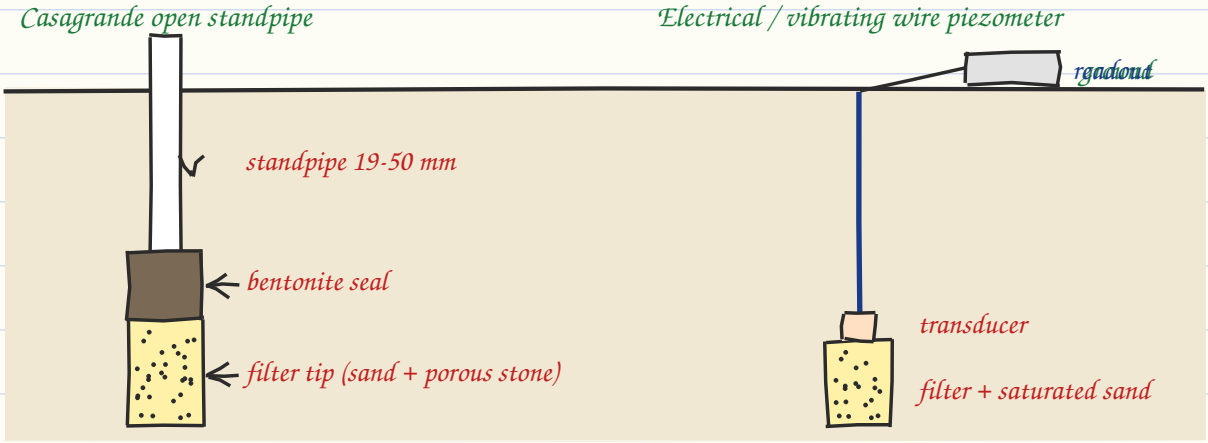


Fig 5.4 Standpipe (Casagrande) and electrical piezometer

Type	Working	Remarks
Open standpipe (Casagrande)	Porous tip connected to a vertical PVC pipe; water level in pipe = piezometric head (dip meter)	Cheap, simple, reliable; slow response (time lag) in clay; max depth limited, freezing issues
Hydraulic (twin-tube)	Two de-aired tubes lead to gauge/panel	Good for fills; needs de-airing
Pneumatic	Diaphragm closes at tip; gas pressure applied until balance - open vent	No lag, remote reading; for earth dams
Vibrating wire	Diaphragm + wire frequency	Excellent long-term, automatic data logging
Electrical (strain-gauge)	Pressure transducer	Quick, but drift

- Installation: borehole -> sand pocket around filter tip (0.3-0.5 m) -> bentonite plug above -> grout/ backfill. Test response (falling-head) and saturate the tip.
- Hydrostatic head $h = u/\gamma_w$; piezometric level compared with elevation gives flow direction. Hvorslev time-lag $T = A/(F\,k)$; where F shape factor.
- Use in clay with slow response: choose diaphragm types.

8. Pressure cells (earth pressure cells)

Measure total stress in soil (or contact pressure between soil and structure: retaining walls, tunnel lining, bridge abutments, dam core). Effective stress = total - pore pressure (from piezometer).

- Construction: two circular stainless steel plates welded at the edge, space between them filled with de-aired oil/mercury; pressure measured by a pneumatic or vibrating-wire transducer (diaphragm type).
- Types: (i) Embedded (total pressure) cells - in soil mass, (ii) Contact cells - flush on structure face, (iii) Hydraulic, pneumatic, vibrating wire etc.
- Installation: cell placed in thin sand/fine soil cushion, orientation noted; stiffness of cell should be similar to soil to avoid arching / over-registration; aspect ratio (diameter: thickness) at least 10 : 1.
- Errors: stress concentration, temperature, installation disturbance, cell/soil stiffness ratio; calibrate in the lab.

9. Inclinometers

Measure lateral (horizontal) deformation vs depth - slopes, landslides, retaining walls, diaphragm walls, embankment toe, piles. Also called slope indicator.

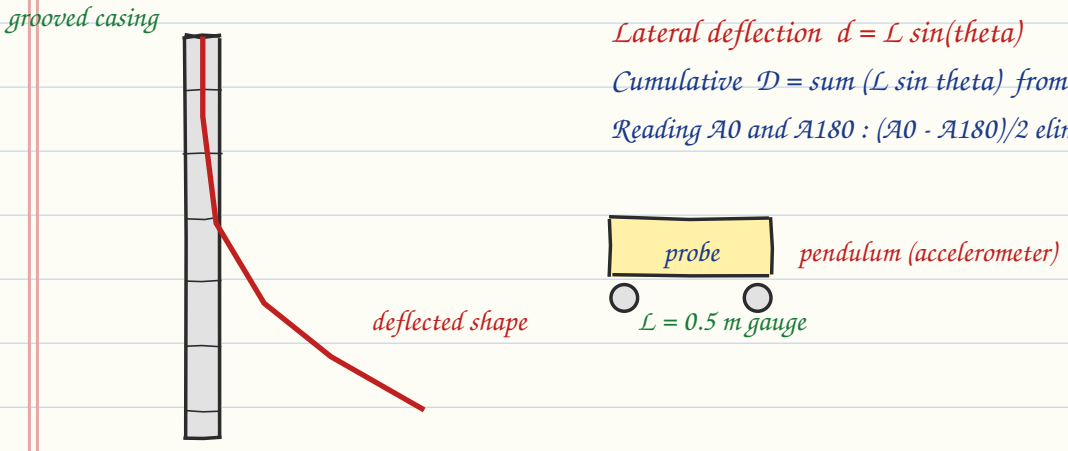


Fig 5.5 Inclinometer casing and probe

- Components: grooved plastic/aluminium casing installed in a borehole with grout, the bottom fixed in stable stratum; probe (torpedo) with two wheels and pendulum/MEMS accelerometer (servo), cable with depth marks, and readout.
- Procedure: lower probe to bottom, take readings at every 0.5 m going up (A0 and A180 face), repeat in perpendicular (B-axis). Compare with the initial reading (baseline).
- Tilt q gives lateral deflection per gauge length: $d = L \sin q$; cumulative deflection at depth z : $D = S\, L \sin q$ (from bottom up).

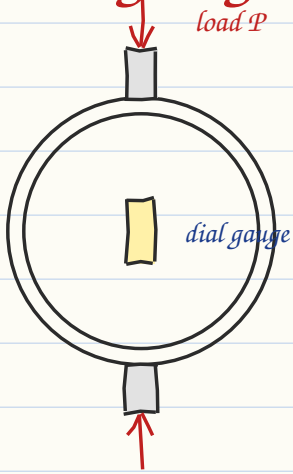
- Output: deflection vs depth, rate of movement, shear zone (sudden change of slope) - slip surface depth.

> Example. Gauge length 0.5 m; readings sin q at successive 0.5 m intervals (from base): 0, 0.002, 0.004, 0.010, 0.016. Deflection at top = 0.5 × (0 + 0.002 + 0.004 + 0.010 + 0.016) = 0.5 × 0.032 = 0.016 m = 16 mm. The largest increments occur between the 3rd-5th points -> shear zone.

Tiltmeter

Measures rotation of a structure/ground (arc-seconds): used for monitoring wall rotation, building tilt, dam crest, adjacent structures during tunnelling.

10. Proving ring



Proving ring: elastic steel ring + dial gauge

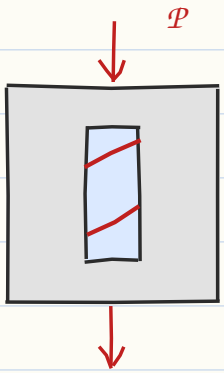
- Load P = K × (dial reading - zero)
- K = calibration constant (N per division)
- Capacity 5 kN - 2000 kN
- Used for CBR, UCS, triaxial, vane
- Calibrated against standard dead weights
- Accuracy about 0.1 - 1 %

Fig 5.6 Proving ring

A precisely machined steel ring with a dial gauge (or LVDT) across its internal diameter. When load is applied along the diameter, the ring deflects elastically and the dial reading is proportional to load. Load P = K × (dial reading) where K is the ring constant from calibration (N/div.).

- Calibration: load with standard weights / compression machine; plot load vs dial reading; check linearity; re-calibrate every year.
- Used in laboratory (UCS, CBR, triaxial, direct shear machines) and also in field for measuring jack loads in plate load tests, pile load tests.
- Advantages: simple, robust, no power; Limitations: manual reading, temperature effect, sensitive to shock.

11. Load cells



Strain gauge load cell

- Steel column with 4 gauges (2 axial, 2 transverse)
- Wheatstone bridge: Vout/Vin ~ GF x e (full bridge)
- Hydraulic load cell: fluid pressure in diaphragm
- Vibrating wire load cell: P = k (f0^2 - f^2)
- Uses: anchor / strut load, pile test, tie rods
- Capacity 10 kN to 10 MN

Fig 5.7 Load cells

Type	Principle	Application
Strain gauge (electrical)	Gauges on elastic steel member; output in mV	Tie-back anchors, struts, pile test
Hydraulic	Load -> pressure in fluid in diaphragm, read by Bourdon gauge/transducer	Heavy loads, jack monitoring
Vibrating wire	Wire frequency in steel body	Long-term monitoring, rock anchors, ground anchors
Pneumatic	Air pressure balances load	Hazardous areas (no electricity)
Photo-elastic	Stress-induced optical change	Rare / research

- Capacity: 10 kN to 10 MN; accuracy 0.5 - 1 % of full scale; compensated for temperature; centrally loaded with bearing plates.
- Calibrate with a testing machine; protect from water (waterproof sealing).

12. Displacement gauges

Gauge	Use
Dial gauge (mechanical)	Settlement in plate load / consolidation tests (0.01 mm resolution, range 25 - 50 mm)
LVDT (linear variable differential transformer)	Electrical displacement transducer; core moves in coils; automatic data acquisition; excellent for lab & field
Extensometer (rod / tape / borehole multi-point)	Relative movement along a borehole; tunnel convergence; rock mass deformation

Gauge	Use
Settlement gauge (magnet-extensometer, hydrostatic profile gauge, plate)	Vertical settlement of fills/embankments at various depth
Crack meters / joint meters	Opening of cracks and joints in structures/ rock
Surveying (total station, GNSS, levelling)	Surface monuments - absolute movement
Fibre optic & Tilt sensors	Distributed strain / deformation

Selection rules for any instrument: (1) Why measure? Define the questions first; (2) predict expected values; (3) choose reliable, accurate, robust instrument with required range; (4) plan calibration, installation, protection and baseline readings; (5) plan data collection, logging, thresholds & action levels; (6) review data continuously.

13. Summary of instruments

Instrument	Measures	Unit	Typical site
Strain gauge	Strain in steel/concrete	microstrain	Piles, struts, tunnel
Piezometer	Pore water pressure	kPa / m head	Dam, slopes, excavation
Earth pressure cell	Total stress	kPa	Retaining wall, embankment
Inclinometer	Lateral deflection	mm	Landslide, diaphragm wall
Proving ring	Load (force)	kN	Lab/field loading tests
Load cell	Load in anchor/strut	kN / MN	Anchors, piles
Displacement gauge	Movement	mm	Settlement, crack

14. Important questions

- What are the contents of a soil exploration report? Explain each briefly.

- Write about the identification and calculations needed in a soil report. Include an example for safe bearing capacity.
- Explain the working of different types of piezometers. Which is best for clay?
- Write short notes on strain gauges and vibrating wire gauges.
- Describe the inclinometer and how lateral displacement is computed.
- Explain the proving ring and its calibration.
- Write short notes on load cells and displacement gauges.
- Why is instrumentation needed in an earth dam? Name suitable instruments.